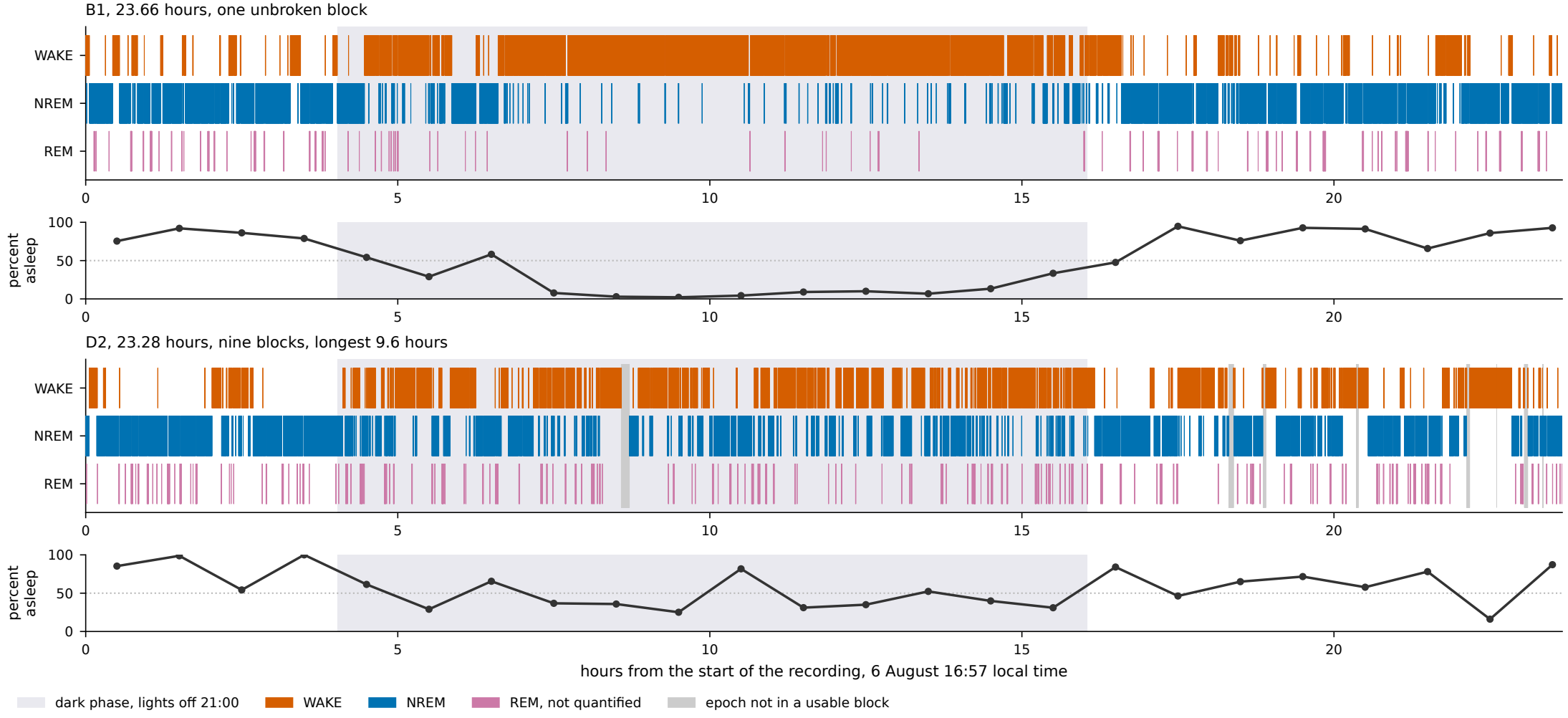
Figure 5. Sleep and wake across a full light and dark cycle, the two animals that worked



Automatic staging on 4 second epochs, drawn as state bands rather than a line, because a line through ordered state codes draws a transition from wake to REM as though it passed through NREM. Epochs outside a usable block are drawn in grey rather than joined over. Wake is separated from sleep by an Otsu threshold on muscle amplitude, refit per animal; see figure 6 for how much that threshold matters. REM is shown for completeness and MUST NOT be reported as a percentage: two independent rules disagree by a factor of five on this montage. Nothing in the quality gate, the staging rule or the persistence statistic has any knowledge of the clock, so the light and dark contrast is external to the method.